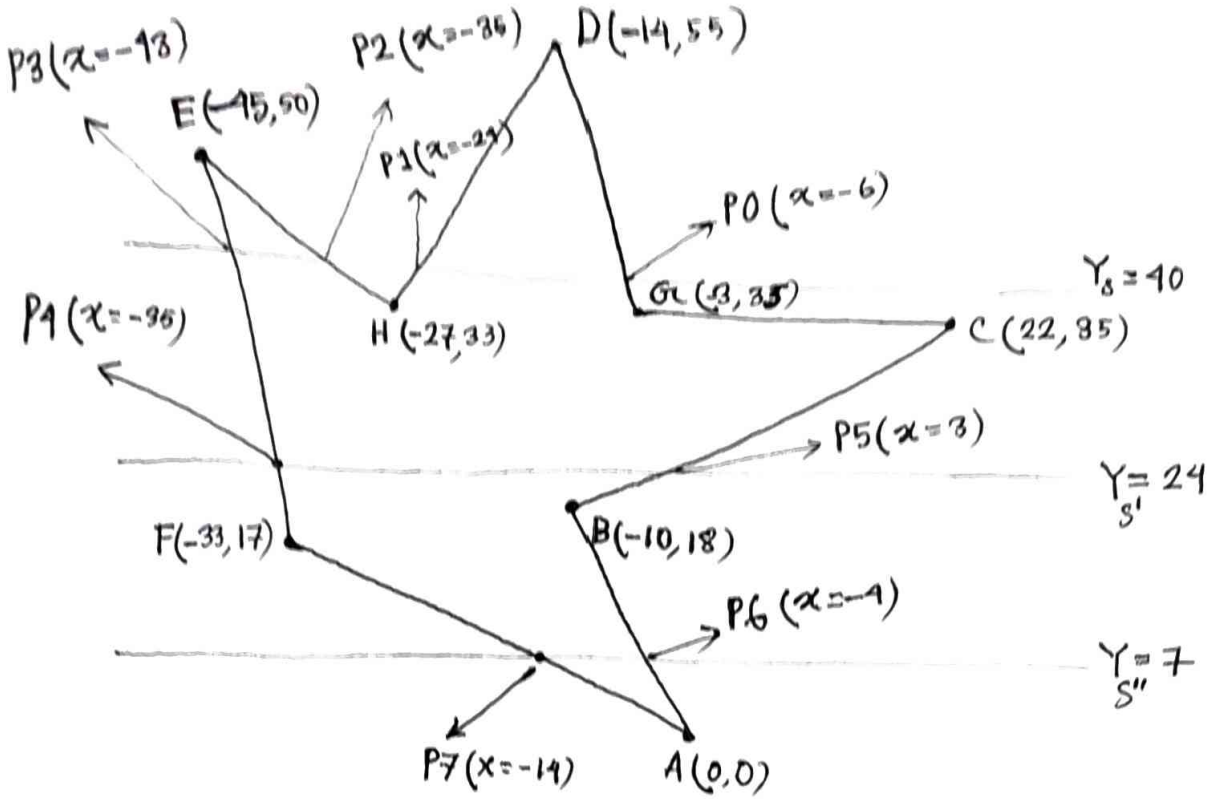
CSE423: Fall'24

Assignment-01

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My student ID: 22101671 . A = 22, B = 10, C = 16, D = 71

We get:

Vertex	Color (RGB)
A	(22, 10, 16)
B	(22, 10, 71)
C	(22, 16, 10)
D	(22, 16, 71)
E	(22, 71, 10)
F	(22, 71, 16)
Gr	(10, 22, 16)
H	(71, 10, 16)

P0 Calculations:

$$P0 = Gr + (D - Gr) \times \frac{Y_{s'} - Gr_y}{D_y - Gr_y}$$

$$= (10,22,16) + \{(22,16,71) - (10,22,16)\} \times \frac{40-35}{55-35}$$

$$= (13, 20.5, 29.75)$$

$$\approx (0.051, 0.080, 0.117)$$

P1 calculations:

$$P1 = H + (D - H) \times \frac{Y_s - H_y}{D_y - H_y}$$

$$= (71,10,16) + \{(22,16,71) - (71,10,16)\} \times \frac{40-33}{55-33}$$

$$= (55.41, 11.91, 33.50)$$

$$\approx (0.217, 0.047, 0.131)$$

P2:

$$P2 = H + (E-H) \times \frac{x_{P2} - x_H}{x_E - x_H}$$

$$= (71,10,16) + \{(22,71,10) - (71,10,16)\} \times \frac{-35-(-27)}{-45-(-27)}$$

$$= (49.22, 37.11, 13.33)$$

$$\approx (0.193, 0.146, 0.052)$$

P3:

$$P3 = F + (E-F) \times \frac{Y_S - Y_F}{Y_E - Y_F}$$

$$= (22,71,16) + \{(22,71,10) - (22,71,16)\} \times \frac{40-17}{50-17}$$

$$= (22,71,11.82)$$

$$\approx (0.086, 0.278, 0.046)$$

P4:

$$P4 = F + (E-F) \times \frac{Y_{S'} - Y_F}{Y_E - Y_F}$$

$$= (22,71,16) + \{(22,71,10) - (22,71,16)\} \times \frac{24-17}{50-17}$$

$$= (22,71,14.73)$$

$$\approx (0.086, 0.278, 0.058)$$

P5:

$$P5 = B + (C-B) \times \frac{x_{P5} - x_B}{x_C - x_B}$$

$$= (22,10,71) + \{(22,16,10) - (22,10,71)\} \times \frac{3-(-10)}{22-(-10)}$$

$$= (22, 12.44, 46.22)$$

$$\approx (0.086, 0.049, 0.181)$$

P6:

$$P6 = A + (B-A) \times \frac{Y_{S''} - Y_A}{Y_B - Y_A}$$

$$= (22,10,16) + \{(22,10,71) - (22,10,16)\} \times \frac{7-0}{18-0}$$

$$= (22, 10, 37.39)$$

$$\approx (0.086, 0.039, 0.147)$$

P7:

$$P7 = A + (F-A) \times \frac{x_{P7} - x_A}{x_F - x_A}$$

$$= (22,10,16) + \{(22,71,16) - (22,10,16)\} \times \frac{-14-0}{-33-0}$$

$$= (22, 35.88, 16)$$

$$\approx (0.086, 0.141, 0.063)$$

RGB → HSV



Point P	R	G	B	C_{max}	Hue	Saturation	Value
P0	0.051	0.080	0.117	B	$\left(4+\frac{0.051-0.080}{0.117-0.051}\right)\times 60^\circ = 213.63^\circ$	$1-\frac{0.051}{0.117}$ $=0.564$	0.051
P1	0.217	0.047	0.131	R	$\left(0+\frac{0.047-0.131}{0.217-0.047}\right)\times 60^\circ$ $=-29.65^\circ+360^\circ$ $=330.35^\circ$	$1-\frac{0.047}{0.217}$ $=0.783$	0.047
P2	0.193	0.146	0.052	R	$\left(0+\frac{0.146-0.052}{0.193-0.052}\right)\times 60^\circ$ $=40^\circ$	$1-\frac{0.052}{0.193}$ $=0.731$	0.052
P3	0.086	0.278	0.046	G	$\left(2+\frac{0.046-0.086}{0.278-0.046}\right)\times 60^\circ$ $=109.66^\circ$	$1-\frac{0.046}{0.278}$ $=0.835$	0.046
P4	0.086	0.278	0.058	G	$\left(2+\frac{0.058-0.086}{0.278-0.056}\right)\times 60^\circ$ $=112.43^\circ$	$1-\frac{0.056}{0.278}$ $=0.799$	0.056
P5	0.086	0.049	0.181	B	$\left(4+\frac{0.086-0.049}{0.181-0.049}\right)\times 60^\circ$ $=256.82^\circ$	$1-\frac{0.049}{0.181}$ $=0.729$	0.049
P6	0.086	0.039	0.147	B	$\left(4+\frac{0.086-0.039}{0.147-0.039}\right)\times 60^\circ$ $=266.11^\circ$	$1-\frac{0.039}{0.147}$ $=0.735$	0.039
P7	0.086	0.141	0.063	G	$\left(2+\frac{0.063-0.086}{0.141-0.063}\right)\times 60^\circ$ $=102.31^\circ$	$1-\frac{0.063}{0.141}$ $=0.553$	0.063